

RUGENX

(PVT) LTD.

ENGINEERED TO PERFORM

Engineering Tomorrow's Industrial Solutions

Engineering Consultancy | Product Development | R&D | Simulation | Automation | Industrial Solutions

COMPANY OVERVIEW

Engineering Consultancy · Product Development · R&D · CAE Simulation · Industrial Automation · Manufacturing Support

Confidential — Prepared for Client & Partner Review

01 COMPANY INTRODUCTION

About RUGENX

RUGENX Pvt Ltd is an engineering and industrial technology solutions company dedicated to transforming ideas, technical challenges, and industrial requirements into practical, reliable, and production-ready engineering outcomes.

We combine engineering design, advanced simulation, research and development, automation, prototyping, and manufacturing expertise to deliver solutions that measurably improve productivity, efficiency, quality, and operational performance.

Our approach extends beyond conventional engineering consultancy. We partner with clients from the earliest stages of an idea through engineering development, validation, implementation, and continuous improvement functioning as a technical extension of our clients' own teams.

We believe effective engineering should not only solve today's operational problems but also create the foundation for smarter, more sustainable industrial systems tomorrow.

02 OUR VISION

To become a leading engineering and industrial technology solutions company, delivering innovative, reliable, and sustainable solutions for the industries of tomorrow.

We aim to build a strong engineering ecosystem in which technology, research, and practical industrial knowledge converge to create meaningful, lasting value for businesses and society.

03 OUR MISSION

To transform industrial challenges and ideas into practical engineering solutions through innovation, research, advanced technology, and engineering excellence.

We are committed to:

- Developing practical and cost-effective engineering solutions
- Supporting businesses through technology and innovation
- Bridging the gap between engineering concepts and industrial implementation
- Providing accessible R&D capabilities to businesses of all sizes
- Improving industrial productivity and efficiency
- Developing locally relevant engineering technologies
- Supporting sustainable manufacturing and resource utilization

04 WHAT WE DO

RUGENX provides integrated engineering services covering the complete development cycle of industrial products, systems, and processes — from first concept through full-scale implementation.

Our Capabilities

Capability	Description
Engineering Consultancy	Technical analysis, problem solving, system development, and engineering decision support.
Product Development	Transforming concepts into engineered products and production-ready systems.
Research & Development	Developing new technologies, machinery, processes, and customised engineering solutions.
CAE & Simulation	Using engineering simulation to evaluate and optimise designs before physical manufacturing.
Industrial Automation	Developing automated systems to improve productivity, consistency, monitoring, and control.
Product & Process Optimisation	Improving existing products, machinery, and manufacturing processes through engineering analysis.
Prototyping & Testing	Developing prototypes and validating engineering concepts through practical testing.
Manufacturing Support	Supporting fabrication, assembly, commissioning, troubleshooting, and production implementation.

05 ENGINEERING SERVICES — PRODUCT DESIGN & DEVELOPMENT

We develop mechanical and industrial products from initial concepts through detailed engineering.

Services Include

- Concept development
- Product architecture
- Mechanical design
- 3D CAD modelling
- Engineering drawings
- Design optimisation
- Material selection
- Component selection
- Design for Manufacturing (DFM)
- Design for Assembly (DFA)
- Prototype development

Our objective is to create products that are technically sound, manufacturable, maintainable, and commercially practical.

06 CAE & ENGINEERING SIMULATION

Computer-Aided Engineering allows us to evaluate designs before committing significant resources to manufacturing, reducing risk and accelerating development.

CFD — Computational Fluid Dynamics

Used to analyse:

- Airflow
- Heat transfer
- Fluid flow
- Pressure distribution
- Ventilation
- Thermal systems
- Process flow

FEA — Finite Element Analysis

Used to evaluate:

- Structural strength
- Stress distribution
- Deformation
- Thermal behaviour
- Component performance
- Design safety

Benefits

DESIGN → SIMULATE → OPTIMISE → MANUFACTURE

Simulation helps reduce development time, minimise physical prototypes, identify potential failure modes early, and improve overall product performance.

07 INDUSTRIAL AUTOMATION

RUGENX develops customised automation solutions for industrial and commercial applications.

Areas of Focus

- Process automation
- Machine automation
- Sensor-based monitoring
- Data acquisition
- Control systems
- Industrial IoT
- Automated machinery
- Remote monitoring
- Production monitoring
- Smart control systems

We integrate mechanical engineering with electronics, sensors, controllers, software, and data systems to create intelligent, connected engineering solutions.

08 RESEARCH & DEVELOPMENT

RUGENX provides R&D capabilities for organisations that need to develop new products, technologies, processes, or systems.

We can function as an external R&D partner, giving organisations access to engineering development capabilities without the overhead of maintaining a complete internal R&D team.

R&D Activities Include

- Technical feasibility studies
- Concept development
- Experimental development
- Prototype development
- Engineering testing
- Process development
- Technology integration
- Design optimisation
- Performance evaluation
- Product improvement

09 PROCESS OPTIMISATION

Existing systems can often achieve significantly better performance through focused engineering analysis and optimisation.

RUGENX Helps Organizations Improve

- Productivity
- Energy efficiency
- Material utilisation
- Production time
- Product quality
- Equipment performance
- Process reliability
- Waste generation
- Operational costs

Our approach combines engineering analysis, process understanding, data, simulation, and practical implementation.

10 PROTOTYPING & TESTING

A successful engineering solution must perform reliably in the real world. RUGENX supports the transition from digital engineering models to physical prototypes and functional systems.

Prototyping Capabilities

- Functional prototypes
- Mechanical assemblies
- Sheet-metal systems
- Machine prototypes
- Automated systems
- Test rigs
- Experimental equipment

Testing May Include

- Functional testing
- Performance testing
- Thermal testing
- Structural testing
- Process testing
- Operational validation

11 MANUFACTURING SUPPORT

RUGENX supports clients during the transition from engineering design to manufacturing and implementation.

Manufacturing Support Includes

- Fabrication-ready engineering drawings
- Manufacturing documentation
- Component specifications
- Material selection
- Manufacturing method selection
- Supplier technical coordination
- Assembly support
- Installation support
- Commissioning assistance
- Troubleshooting
- Design modifications

Our goal is to ensure that an engineering design is not only technically correct but also practical to manufacture and operate.

12 INDUSTRIES WE SERVE

RUGENX develops solutions for a wide range of industrial and commercial applications.

Industry	Focus
Manufacturing	Machinery, production systems, process optimisation, automation, and engineering support.
Food & Agro Processing	Processing machinery, drying systems, grinding systems, automation, and process improvement.
Waste Management & Recycling	Recycling systems, material processing equipment, process development, and resource recovery technologies.
Industrial Automation	Automated machinery, sensor systems, monitoring, control, and smart industrial solutions.
Energy & Thermal Systems	Thermal equipment, heat transfer systems, energy optimisation, and alternative energy applications.
Automotive & Transportation	Engineering analysis, product development, mechanical systems, and technology integration.
SMEs & Startups	Technical product development, prototype development, engineering consultancy, and external R&D support.

13 TECHNOLOGY & ENGINEERING CAPABILITIES

RUGENX integrates modern engineering technologies throughout the product and system development lifecycle.

Digital Engineering

- 3D CAD
- Mechanical design
- Engineering drawings
- CAD/CAM
- Digital prototyping

Simulation

- CFD
- FEA
- Thermal analysis
- Structural analysis
- Flow analysis

Automation

- Sensors
- Microcontrollers
- PLC-based systems
- Data acquisition
- Industrial IoT
- Monitoring systems

Manufacturing

- Fabrication
- Machining
- Sheet-metal manufacturing
- Assembly
- Prototype manufacturing

14 OUR BUSINESS MODEL

RUGENX works with clients through multiple engagement models, chosen to fit project requirements.

Engagement Model	Description
Project-Based Engineering	Complete engineering solutions developed for specific technical requirements.
Engineering Consultancy	Technical expertise provided to solve engineering problems and support decision making.
External R&D Partnership	Long-term technical collaboration where RUGENX functions as an external engineering and R&D team.
Product Development Partnership	Collaborative development of new products from concept to commercialisation.
Technology Integration	Integration of mechanical, electronic, software, automation, and data technologies into practical systems.
Manufacturing & Implementation Support	Engineering support throughout fabrication, installation, commissioning, and production.

15 SELECTED DEVELOPMENT AREAS

RUGENX is building capabilities across several strategic technology areas.

Area	Description
Automated Processing Systems	Development of automated systems for food and agricultural processing.
Drying & Dehydration Technologies	Engineering of controlled drying systems with temperature, humidity, time, and process monitoring.
Grinding & Powder Processing	Development of grinding, milling, material handling, collection, and packaging systems.
Waste Plastic Processing	Development and integration of systems for plastic sorting, processing, recycling, and material recovery.
Industrial Monitoring	Sensor-based systems for monitoring industrial parameters and equipment performance.
Custom Machinery	Design and development of machinery according to specific industrial requirements.

20 OUR FUTURE

RUGENX is building toward a future where locally developed engineering technologies can compete with global solutions.

Our Long-Term Direction Includes Expansion Into

- Advanced industrial automation
- Smart manufacturing
- Industrial IoT
- Advanced product development
- Engineering simulation
- Sustainable technologies
- Recycling technologies
- Process automation
- Industrial machinery
- Technology commercialisation

We aim to develop RUGENX into a platform where engineering ideas can move efficiently from concept to commercial industrial application.

21 OUR COMMITMENT

At RUGENX, we believe engineering should create measurable value. Every project is an opportunity to:

**SOLVE A PROBLEM · IMPROVE A PROCESS · DEVELOP A TECHNOLOGY · CREATE A PRODUCT ·
BUILD A BETTER FUTURE**

We are committed to maintaining a culture of engineering excellence, innovation, practical problem solving, continuous learning, and responsible technology development.

24 CONTACT

RUGENX PVT LTD

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Website: [Company Website]

Email: Rugenxinnovations@gmail.com

Telephone: +94 74 18 500 60

Address: 563/C, Nawagamuwa South, Ranala.

Let's engineer the future together.